

Supplementary Table 9. Reviewer-requested sensitivity analyses for development-only model selection, SSL PCA dimensionality reduction, elastic-net regularization and pH-threshold outcome definitions

Frontiers in Signal Processing R1 supplementary table

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Supplementary Table 9. Reviewer-requested sensitivity analyses for development-only model selection, SSL PCA dimensionality reduction, elastic-net regularization and pH-threshold outcome definitions

Standalone display version for LaTeX upload/review. Machine-readable source: `source_tables/Supplementary_Table_9_reviewer_sensitivity_analyses.csv`. Rows: 24; columns: 24. The source table is split below into source-defined sections and column blocks so that all non-empty cells remain readable.

analysis type: Development model selection

Table 1: Supplementary Table 9, Development model selection, column block 1
of 3

config	model	selection split	Development train records	Development validation records
p80 d64 l3 channel cw02	Signal + SSL	75% original-training records fit / 25% original-training records validation	289	97
p80 d64 l3 channel cw02	Signal + SSL + phenotype	75% original-training records fit / 25% original-training records validation	289	97
p60 d64 l2 mixed cw02	Signal + SSL	75% original-training records fit / 25% original-training records validation	289	97
p60 d64 l2 mixed cw02	Signal + SSL + phenotype	75% original-training records fit / 25% original-training records validation	289	97
p80 d48 l2 mixed cw01	Signal + SSL	75% original-training records fit / 25% original-training records validation	289	97
p80 d48 l2 mixed cw01	Signal + SSL + phenotype	75% original-training records fit / 25% original-training records validation	289	97
p120 d64 l2 block cw02	Signal + SSL	75% original-training records fit / 25% original-training records validation	289	97
p120 d64 l2 block cw02	Signal + SSL + phenotype	75% original-training records fit / 25% original-training records validation	289	97

Table 2: Supplementary Table 9, Development model selection, column block 2
of 3

config	model	Development train events	Development validation events	AUROC
p80 d64 l3 channel cw02	Signal + SSL	59	20	0.7078
p80 d64 l3 channel cw02	Signal + SSL + phenotype	59	20	0.7045
p60 d64 l2 mixed cw02	Signal + SSL	59	20	0.6942
p60 d64 l2 mixed cw02	Signal + SSL + phenotype	59	20	0.6981
p80 d48 l2 mixed cw01	Signal + SSL	59	20	0.7396
p80 d48 l2 mixed cw01	Signal + SSL + phenotype	59	20	0.75
p120 d64 l2 block cw02	Signal + SSL	59	20	0.6519
p120 d64 l2 block cw02	Signal + SSL + phenotype	59	20	0.6597

Table 3: Supplementary Table 9, Development model selection, column block 3
of 3

config	model	AUPRC	Brier score	event prevalence
p80 d64 l3 channel cw02	Signal + SSL	0.3569	0.2411	0.2062
p80 d64 l3 channel cw02	Signal + SSL + phenotype	0.3569	0.2414	0.2062
p60 d64 l2 mixed cw02	Signal + SSL	0.3415	0.2282	0.2062
p60 d64 l2 mixed cw02	Signal + SSL + phenotype	0.3564	0.2287	0.2062
p80 d48 l2 mixed cw01	Signal + SSL	0.4291	0.2277	0.2062
p80 d48 l2 mixed cw01	Signal + SSL + phenotype	0.4482	0.2267	0.2062
p120 d64 l2 block cw02	Signal + SSL	0.2925	0.2495	0.2062
p120 d64 l2 block cw02	Signal + SSL + phenotype	0.3	0.2467	0.2062

analysis type: development selected test evaluation

Table 4: Supplementary Table 9, development selected test evaluation, column
block 1 of 3

config	model	selection split	AUROC	AUPRC
p80 d48 l2 mixed cw01	Signal + SSL + phenotype	selected by original-training internal validation; original held-out test evaluated once	0.6348	0.3872

Table 5: Supplementary Table 9, development selected test evaluation, column
block 2 of 3

config	model	Brier score	event prevalence	Train records
p80 d48 l2 mixed cw01	Signal + SSL + phenotype	0.2542	0.2048	386

Table 6: Supplementary Table 9, development selected test evaluation, column
block 3 of 3

config	model	Held-out test records	Train events	Held-out test events
p80 d48 l2 mixed cw01	Signal + SSL + phenotype	166	79	34

analysis type: dimensionality regularization

Table 7: Supplementary Table 9, dimensionality regularization, column block 1 of 4

model	AUROC	AUPRC	Brier score	event prevalence
signal + ssl64 + signal phenotype l2	0.6168	0.4016	0.2652	0.2048
Signal + SSL pca5 + signal phenotype l2	0.6286	0.3863	0.2547	0.2048
Signal + SSL pca5 + signal phenotype elasticnet	0.6279	0.3801	0.2548	0.2048
Signal + SSL pca10 + signal phenotype l2	0.6212	0.4241	0.2588	0.2048
Signal + SSL pca10 + signal phenotype elasticnet	0.6194	0.4026	0.2584	0.2048
Signal + SSL pca20 + signal phenotype l2	0.6535	0.4256	0.2566	0.2048
Signal + SSL pca20 + signal phenotype elasticnet	0.6549	0.4172	0.2557	0.2048

Table 8: Supplementary Table 9, dimensionality regularization, column block 2
of 4

model	AUROC	Train records	Held-out test records	Train events
signal + ssl64 + signal phenotype l2	0.6168	386	166	79
Signal + SSL pca5 + signal phenotype l2	0.6286	386	166	79
Signal + SSL pca5 + signal phenotype elasticnet	0.6279	386	166	79
Signal + SSL pca10 + signal phenotype l2	0.6212	386	166	79
Signal + SSL pca10 + signal phenotype elasticnet	0.6194	386	166	79
Signal + SSL pca20 + signal phenotype l2	0.6535	386	166	79
Signal + SSL pca20 + signal phenotype elasticnet	0.6549	386	166	79

Table 9: Supplementary Table 9, dimensionality regularization, column block 3
of 4

model	AUROC	Held-out test events	SSL dimensions	penalty
signal + ssl64 + signal phenotype l2	0.6168	34	64	l2
Signal + SSL pca5 + signal phenotype l2	0.6286	34	5	l2
Signal + SSL pca5 + signal phenotype elasticnet	0.6279	34	5	elasticnet
Signal + SSL pca10 + signal phenotype l2	0.6212	34	10	l2
Signal + SSL pca10 + signal phenotype elasticnet	0.6194	34	10	elasticnet
Signal + SSL pca20 + signal phenotype l2	0.6535	34	20	l2
Signal + SSL pca20 + signal phenotype elasticnet	0.6549	34	20	elasticnet

Table 10: Supplementary Table 9, dimensionality regularization, column block
4 of 4

model	AUROC	n predictors before one hot	events per predictor before one hot	PCA explained variance ratio
signal + ssl64 + signal phenotype l2	0.6168	77	1.026	
Signal + SSL pca5 + signal phenotype l2	0.6286	18	4.389	0.84
Signal + SSL pca5 + signal phenotype elasticnet	0.6279	18	4.389	0.84
Signal + SSL pca10 + signal phenotype l2	0.6212	23	3.435	0.9567
Signal + SSL pca10 + signal phenotype elasticnet	0.6194	23	3.435	0.9567
Signal + SSL pca20 + signal phenotype l2	0.6535	33	2.394	0.9953
Signal + SSL pca20 + signal phenotype elasticnet	0.6549	33	2.394	0.9953

analysis type: outcome sensitivity

Table 11: Supplementary Table 9, outcome sensitivity, column block 1 of 4

model	AUROC	AUPRC	Brier score	event prevalence
Classical signal features	0.6303	0.3366	0.2525	0.2048
Signal + SSL + phenotype	0.6168	0.4016	0.2652	0.2048
Classical signal features	0.6398	0.3882	0.2525	0.1988
Signal + SSL + phenotype	0.6744	0.4432	0.2489	0.1988
Classical signal features	0.7809	0.4773	0.2183	0.1145
Signal + SSL + phenotype	0.8006	0.487	0.1941	0.1145
Classical signal features	0.8181	0.5409	0.207	0.0904
Signal + SSL + phenotype	0.8631	0.526	0.1726	0.0904

Table 12: Supplementary Table 9, outcome sensitivity, column block 2 of 4

model	AUROC	Train records	Held-out test records	Train events
Classical signal features	0.6303	386	166	79
Signal + SSL + phenotype	0.6168	386	166	79
Classical signal features	0.6398	386	166	72
Signal + SSL + phenotype	0.6744	386	166	72
Classical signal features	0.7809	386	166	37
Signal + SSL + phenotype	0.8006	386	166	37
Classical signal features	0.8181	386	166	25
Signal + SSL + phenotype	0.8631	386	166	25

Table 13: Supplementary Table 9, outcome sensitivity, column block 3 of 4

model	AUROC	Held-out test events	outcome	outcome definition
Classical signal features	0.6303	34	neonatal risk	primary composite: cord pH ≤ 7.15 or 5-minute Apgar ≤ 7
Signal + SSL + phenotype	0.6168	34	neonatal risk	primary composite: cord pH ≤ 7.15 or 5-minute Apgar ≤ 7
Classical signal features	0.6398	33	cord ph lt 7 15	cord pH ≤ 7.15 only
Signal + SSL + phenotype	0.6744	33	cord ph lt 7 15	cord pH ≤ 7.15 only
Classical signal features	0.7809	19	cord ph lt 7 10	cord pH ≤ 7.10 only
Signal + SSL + phenotype	0.8006	19	cord ph lt 7 10	cord pH ≤ 7.10 only
Classical signal features	0.8181	15	cord ph lt 7 05	cord pH ≤ 7.05 only
Signal + SSL + phenotype	0.8631	15	cord ph lt 7 05	cord pH ≤ 7.05 only

Table 14: Supplementary Table 9, outcome sensitivity, column block 4 of 4

model	AUROC	AUPRC / test prevalence
Classical signal features	0.6303	1.643
Signal + SSL + phenotype	0.6168	1.961
Classical signal features	0.6398	1.953
Signal + SSL + phenotype	0.6744	2.229
Classical signal features	0.7809	4.17
Signal + SSL + phenotype	0.8006	4.255
Classical signal features	0.8181	5.985
Signal + SSL + phenotype	0.8631	5.821